



Pearl

△ Spherical pearl showing iridescence

Pearls are natural gems produced by the pearl oyster and the freshwater pearl mussel. Although other kinds of mollusc can produce a “pearl”, these have little value as they are not composed of nacre (the same substance as mother-of-pearl). Nacre is secreted in response to a microscopic irritant in the mollusc’s soft tissue. The concentric rings of nacre around the particle create the particular iridescence of pearls due to the way the overlapping layers diffract light waves. A pearl’s colour is described in terms of body colour and overtone – the most common body colour is white, though it can range widely, and the overtone is the colour that seems to appear only on the surface of the pearl.

Natural and man-made

Natural pearls that form in the wild are rare, and therefore valuable, and pearl divers have to open hundreds of pearl oysters before chancing on a specimen. Although diving for natural pearls still occurs in Bahrain and off the coast of Australia, today’s pearls are largely cultured, making them much more affordable: an artificial nucleus, such as a round shell bead, is placed in the oyster or mussel for the nacre to form around. Freshwater pearls are cheaper because the freshwater mussel can produce up to 20 pearls at a time, whereas the smaller saltwater oyster can create just one. Saltwater pearls are differentiated by region: South Sea pearls are most valued due to their size, Tahitian pearls are next for their colours (black among them), with Akoya the least prized, being the most common.

Key pieces



Roman earring | Crafted in a style common in the 3rd century CE, this extravagant Roman earring is set with garnets cut *en cabochon* and five natural pearls suspended on gold mounts.



The Hope Pearl | One of the former owners of the Hope Diamond (see pp.62–63), 19th-century collector Henry Hope also acquired this stunning 450-carat bronze-to-white baroque pearl capped with gold and enamel.

Specification

Chemical name Calcium carbonate | **Formula** CaCO_3
Colours white, pink, silver, cream, brown, green, blue, black, yellow | **Structure** Amorphous | **Hardness** 2.5–4.5
SG 2.60–2.85 | **RI** 1.52–1.69 | **Lustre** Pearly



Locations

1 Coastal waters of Japan **2** Coastal waters of China
3 Coastal waters of Australia



The Baroda Necklace | Originally a seven-strand necklace belonging to Maharajah Khanda Rao Gaekwad of Baroda, India, this piece was reduced in size in the mid-20th century. It is still the world’s most valuable pearl necklace.